## ANSWERS OF MODEL TEST PAPER 1

## INTERMEDIATE: GROUP – II

## PAPER – 6: FINANCIAL MANAGEMENT & STRATEGIC MANAGEMENT

## PAPER 6A : FINANCIAL MANAGEMENT

## PART I

**1. I. (b) ₹ 35,55,556**

**II. (c) ₹ 30,03,733**

**III. (a) ₹ 8,83,200**

**IV. (d) ₹ 4,83,200**

**V. (a) 16.09%**

**Working Note**

| **Particulars** | **(₹)** |
|---|---|
| Total Sales | ₹ 200 lakhs |
| Credit Sales (80%) | ₹ 160 lakhs |
| Receivables for 40 days | ₹ 80 lakhs |
| Receivables for 120 days | ₹ 80 lakhs |
| Average collection period [(40 x 0.5) + (120 × 0.5)] | 80 days |
| Average level of Receivables (₹ 1,60,00,000 × 80/360) | ₹ 35,55,556 |
| Factoring Commission (₹ 35,55,556 × 2/100) | ₹ 71,111 |
| Factoring Reserve (₹ 35,55,556 × 10/100) | ₹ 3,55,556 |
| Amount available for advance {₹ 35,55,556 - (3,55,556 + 71,111)} | ₹ 31,28,889 |
| Factor will deduct his interest @ 18%: Interest = $\frac{₹31,28,889 \times 18 \times 80}{100 \times 360}$ | ₹ 1,25,156 |
| Advance to be paid (₹ 31,28,889 – ₹ 1,25,156) | ₹ 30,03,733 |

**(i) Statement Showing Evaluation of Factoring Proposal**

| | | ₹ |
|---|---|---|
| A. | **Annual Cost of Factoring to the Company:** | |
| | Factoring commission (₹ 71,111 × 360/80) | 3,20,000 |
| | Interest charges (₹ 1,25,156 × 360/80) | 5,63,200 |
| | Total | 8,83,200 |
| B. | **Company's Savings on taking Factoring Service:** | ₹ |
| | Cost of credit administration saved | 2,40,000 |
| | Bad Debts (₹ 160,00,000 x 1/100) avoided | 1,60,000 |
| | Total | 4,00,000 |

| | | |
|---|---|---|
| C. | Net Cost to the company (A – B) (₹ 8,83,200 – ₹ 4,00,000) | <u>4,83,200</u> |

Effective cost of factoring = $\frac{₹\ 4,83,200}{₹30,03,733} \times 100$ = 16.09%

**2. B. ₹ 3,20,513; ₹ 8.33**

$$\frac{(EBIT - I)(1 - t) - D_p}{N_1} = \frac{(EBIT - I)(1 - t) - D_p}{N_2}$$

$$\frac{(x - 0)(1 - 0.35)}{25,000} = \frac{(x - 1,00,000)(1 - 0.35) - 60,000}{10,000}$$

x = EBIT = ₹ 3,20,513

At EBIT of ₹ 3,20,513, EPS under both options will be the same i.e., ₹ 8.33 per share

**3. D. 1.15**

FL= % change in NP/%change in EBIT=6.9/6=1.15

**4. C. 3 years**

These deposits may be accepted for a period of six months to three years.

## PART II

**1. (a)**

| Particulars | (₹' in lakhs) |
|---|---:|
| Net Profit | 54 |
| Less: Preference dividend | 24 |
| Earnings for equity shareholders | 30 |
| Earnings per share | 30/2 = ₹ 15 |

Let, the dividend per share be D to get share price of ₹ 120.

$$P = \frac{D + \frac{r}{Ke}(E - D)}{K_e}$$

Where,

P = Market price per share.

E = Earnings per share = ₹ 15

D = Dividend per share

R = Return earned on investment = 22%

$K_e$ = Cost of equity capital = 15%

$$120 = \frac{D + \frac{0.22}{0.15}(15 - D)}{0.15}$$

$$18 = \frac{0.15D + 3.3 - 0.22D}{0.15}$$

$$0.07D = 3.3 – 2.7$$

$$D = 8.57$$

$$\text{D/P ratio} = \frac{DPS}{EPS} \times 100 = \frac{8.57}{15} \text{ x } 100 = 57.13\%$$

So, the required dividend pay-out ratio will be = 57.13%

**(b)** Value of AN Ltd. = $\frac{NOI}{K_o} = \frac{₹\ 10,00,000}{20\%} = ₹\ 50,00,000$

**(i) Return on Shares of Mr. R on AN Ltd.**

| Particulars | Amount (₹) |
|---|---|
| Value of the company | 50,00,000 |
| Market value of debt (50% x ₹ 50,00,000) | 25,00,000 |
| Market value of shares (50% x ₹ 50,00,000) | **25,00,000** |
| **Particulars** | **Amount (₹)** |
| Net operating income | 10,00,000 |
| Interest on debt (10% × ₹ 25,00,000) | 2,50,000 |
| Earnings available to shareholders | **7,50,000** |
| Return on 8% shares (8% × ₹ 7,50,000) | **60,000** |

**(ii)** Implied required rate of return on equity of AN Ltd. = $\frac{₹\ 7,50,000}{₹\ 25,00,000}$ = **30%**

**(c)** **ANVY Ltd**

**Balance Sheet as on 31st March, 2023**

| Liabilities | ₹ | Assets | ₹ |
|---|---|---|---|
| Equity share capital | 2,00,000 | Fixed assets | 1,40,000 |
| Current debt | 60,000 | Cash (balancing figure) | 1,00,000 |
| Long term debt | 60,000 | Inventory | 80,000 |
| | 3,20,000 | | 3,20,000 |

**Working Notes**

1. Total debt = 0.60 x Equity share capital = 0.60 × ₹ 2,00,000 = ₹ 1,20,000

   Further, Current debt to total debt = 0.50. So, current debt = 0.50 x ₹1,20,000 = ₹ 60,000,

   Long term debt = ₹1,20,000 - ₹60,000= ₹ 60,000

2. Fixed assets = 0.70 × Equity share Capital = 0.70 × ₹ 2,00,000 = ₹ 1,40,000

3. Total assets to turnover = 2.5 Times: Inventory turnover = 10 Times

Hence, Inventory /Total assets = 2.5/10=1/4, Total assets = ₹ 3,20,000

Therefore Inventory = ₹ 3,20,000/4 = ₹ 80,000

**2. (a) Cash inflows after tax (CFAT)**

| Particular | ₹ |
|---|---:|
| Current production (units per week) | 5,000 units |
| New capacity (units per week) | 15,000 units |
| Demand (units per week) | 10,000 units |
| Increase in sales (units per week) A. | 5,000 units |
| Contribution per unit (₹ 30,000 x 0.10) B. | 3,000 |
| Increase in contribution A x B x 56 | 84 crores |
| Less: Additional fixed cost | 10 crores |
| Increase in profit | 74 crores |
| Less: Tax @ 40% | 29.6 crores |
| Profit after tax | 44.4 crores |

**Tax shield due to depreciation**

| Year | Depreciation (₹ in Crore) | Tax Shield (₹ in Crore) | PV Factor @ 20% | Total Present Value (₹ in Crore) |
|---|---:|---:|---:|---:|
| 1 | 25.00 | 10 | 0.83 | 8.33 |
| 2 | 18.75 | 7.5 | 0.69 | 5.18 |
| 3 | 14.06 | 5.62 | 0.58 | 3.26 |
| 4 | 10.55 | 4.22 | 0.48 | 2.03 |
| 5 | 7.91 | 3.16 | 0.40 | 1.27 |
| Total | | | | 20.07 |

Tax shield on capital loss = (23.73-20.00) x 30% = ₹ 1.12 crores

Net Present Value (NPV)

| Particulars | Year | Cash Flow (₹ in Crores) | PVAF @ 20% | Present Value (₹ in Crores) |
|---|---:|---:|---:|---:|
| Initial Investment | 0 | (100) | 1 | (100) |
| Working capital | 0 | (3) | 1 | (3) |
| Profit after tax | 1-5 | 44.4 | 2.99 | 132.76 |
| Salvage value | 5 | 20 | 0.40 | 8.00 |
| Tax shield on Depreciation | 1-5 | | | 20.07 |
| Tax shield on capital loss | 5 | 1.12 | 0.40 | 0.45 |
| Release of Working Capital | 5 | 3 | 0.40 | 1.20 |
| NPV | | | | 59.47 |

The company is advised to replace the old machine since the NPV of the new machine is positive.

**(b)** **Cut-off Rate:** It is the minimum rate which the management wishes to have from any project. Usually this is based upon the cost of capital. The management gains only if a project gives return of more than the cut - off rate. Therefore, the cut - off rate can be used as the discount rate or the opportunity cost rate.

**3. (a) Working Note:**

Let the rate of Interest on debenture be x

∴ Rate of Interest on loan = 1.4x

$$\therefore k_d \text{ on debentures} = \frac{\text{Int}(1-t)+\frac{RV-NP}{n}}{\frac{RV+NP}{2}}$$

$$= \frac{100x(1-0.30)+\frac{100-98}{4}}{\frac{100+98}{2}}$$

$$= \frac{70x+0.5}{99}$$

∴ $K_d$ on bank loan = 1.4 x (1 – 0.30) = 0.98x

$$K_e = \frac{EPS}{MPS} = \frac{1}{MPS/EPS} = \frac{1}{PE} = \frac{1}{4} = 0.25$$

$K_e$ = 0.25

**Computation of WACC**

| Capital | Amount | Weights | Cost | Product |
|---|---|---|---|---|
| Equity | 20,00,000 | 0.2 | 0.25 | 0.05 |
| Reserves | 30,00,000 | 0.3 | 0.25 | 0.075 |
| Debentures | 30,00,000 | 0.3 | (70x+0.5)/99 | (21x+0.15)/99 |
| Bank Loan | 20,00,000 | 0.2 | 0.98x | 0.196x |
| | 1,00,00,000 | 1 | | 0.125+0.196x $+\frac{21x+0.15}{99}$ |

WACC = 16%

$$\therefore 0.125+0.196x+\frac{21x+0.15}{99} = 0.16$$

∴ 12.375+19.404x+21x+0.15 = (0.16)(99)

∴ 40.404x = 15.84 – 12.525

∴ 40.404x = 3.315

$\therefore$ x = $\frac{3.315}{40.404}$

$\therefore$ x = 8.20%

(i) Rate of interest on debenture = x = 8.20%

(ii) Rate of interest on Bank loan = 1.4x = (1.4)(8.20%) = 11.48%.

**(b)** In dividend price approach, cost of equity capital is computed by dividing the expected dividend by market price per share. This ratio expresses the cost of equity capital in relation to what yield the company should pay to attract investors. It is computed as:

$$K_e = \frac{D_1}{P_0}$$

Where,

$K_e$= Cost of equity

D = Expected dividend (also written as $D_1$)

$P_0$ = Market price of equity (ex- dividend)

**4. (a)** Limitations of Profit Maximisation objective of financial management.

(i) **The term profit is vague. It does not clarify what exactly it means.** It conveys a different meaning to different people. For example, profit may be in short term or long term period; it may be total profit or rate of profit etc.

(ii) **Profit maximisation has to be attempted with a realisation of risks involved.** There is a direct relationship between risk and profit. Many risky propositions yield high profit. Higher the risk, higher is the possibility of profits. If profit maximisation is the only goal, then risk factor is altogether ignored. This implies that finance manager will accept highly risky proposals also, if they give high profits. In practice, however, risk is very important consideration and has to be balanced with the profit objective.

(iii) **Profit maximisation as an objective does not take into account the time pattern of returns.** Proposal A may give a higher amount of profits as compared to proposal B, yet if the returns of proposal A begin to flow say 10 years later, proposal B may be preferred which may have lower overall profit but the returns flow is more early and quick.

(iv) **Profit maximisation as an objective is too narrow.** It fails to take into account the social considerations as also the obligations to various interests of workers, consumers, society, as well as ethical trade practices. If these factors are ignored, a company cannot survive for long. Profit maximization at the cost of social and moral obligations is a short sighted policy.

**(b)** Some common methods of venture capital financing are as follows:

**(i)** **Equity financing:** The venture capital undertakings generally require funds for a longer period but may not be able to provide returns to the investors during the initial stages. Therefore, the venture capital finance is generally provided by way of equity share capital. The equity contribution of venture capital firm does not exceed 49% of the total equity capital of venture capital undertakings so that the effective control and ownership remains with the entrepreneur.

**(ii)** **Conditional loan:** A conditional loan is repayable in the form of a royalty after the venture is able to generate sales. No interest is paid on such loans. In India venture capital financiers charge royalty ranging between 2 and 15 per cent; actual rate depends on other factors of the venture such as gestation period, cash flow patterns, risk and other factors of the enterprise. Some Venture capital financiers give a choice to the enterprise of paying a high rate of interest (which could be well above 20 per cent) instead of royalty on sales once it becomes commercially sound.

**(iii)** **Income note:** It is a hybrid security which combines the features of both conventional loan and conditional loan. The entrepreneur has to pay both interest and royalty on sales but at substantially low rates. IDBI's VCF provides funding equal to 80 – 87.50% of the projects cost for commercial application of indigenous technology.

**(iv)** **Participating debenture:** Such security carries charges in three phases — in the start-up phase no interest is charged, next stage a low rate of interest is charged up to a particular level of operation, after that, a high rate of interest is required to be paid.

**(c)** **Optimum Capital Structure:** The capital structure is said to be optimum when the firm has selected such a combination of equity and debt so that the wealth of firm is maximum. At this capital structure, the cost of capital is minimum and the market price per share i.e. value of the firm is maximum.

**OR**

Financial leverage indicates the use of funds with fixed cost like long term debts and preference share capital along with equity share capital which is known as trading on equity. The basic aim of financial leverage is to increase the earnings available to equity shareholders using fixed cost fund.

A firm is known to have a positive/favourable leverage when its earnings are more than the cost of debt. If earnings are equal to or less than cost of debt, it will be an negative/unfavourable leverage. When the quantity of fixed cost fund is relatively high in comparison to equity capital it is said that the firm is **‘’trading on equity”.**